

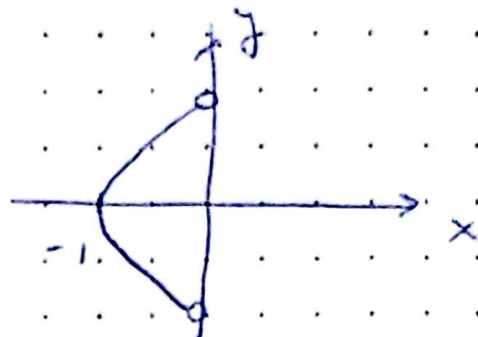
$$1) D_F = \{(x, y) \in \mathbb{R}^2 / |x| \neq 1\} \quad x \neq -1 \wedge y \neq 1$$

$C_0(F)$ el. cerrado

$$\left. \begin{array}{l} C_+(F) = C_-(F) \\ C_-(F) = C_+(F) \end{array} \right\} \text{ del otro lado}$$

$$C_+(F) : x = y^2 - 1, -1 \leq y \leq 1$$

$$\sigma : (-1, 1) \rightarrow \mathbb{R}^2 / \sigma(t) = (t^2 - 1, t)$$



$$2) F = \cos(y - 2x) + e^{2-xy} + xy^2 + 4x$$

$$\nabla F = (\sin(y - 2x) - y e^{2-xy} + y^2, -\sin(y - 2x) - x e^{2-xy} + 2xy)$$

$$\nabla F(1, 2) = (0 - 2 + 4, 0 - 1 + 4) = (2, 3)$$

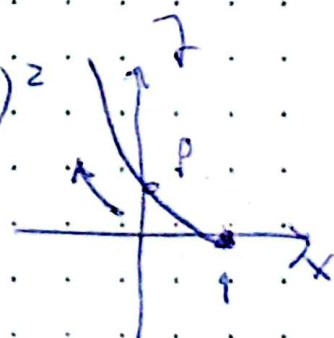
$$\|\nabla F(1, 2)\| = \sqrt{4 + 9} = \sqrt{13} \quad \vec{N}_1 = \left(\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}} \right)$$

$$\vec{N}_2 = \left(\frac{-3}{\sqrt{13}}, \frac{2}{\sqrt{13}} \right) \quad \vec{N}_3 = \left(\frac{3}{\sqrt{13}}, \frac{-2}{\sqrt{13}} \right)$$

$$3) \begin{cases} x = 1 - t^2 \rightarrow t^2 = 1 - x \\ y = t^4 \rightarrow y = (1 - x)^2 = (x - 1)^2 \end{cases}$$

$$t^2 = 1 - x \geq 0 \Rightarrow 1 - x \geq 0 \Rightarrow x \leq 1$$

$$\text{E.C. } y = (x - 1)^2, x \leq 1$$



$$P = (0, 1) \text{ cuando } t / \begin{cases} 1 - t^2 = 0 \\ t^4 = 1 \end{cases} \quad t = 1 \quad t = -1 \neq 0$$

$$f' = (-2t, 4t^3) \rightarrow f'(1) = (-2, 4) \quad \|f'\| = \sqrt{20}$$

$$4) P = (-1, 2, 0) \quad \sigma : [0, 2\pi] \rightarrow \mathbb{R}^3 / \sigma(t) = (1 - 2\cos t, 2\cos t, 2\sin t)$$

$$\text{Ent } t = 1, \text{ sent } t = 0 \quad t = 0$$



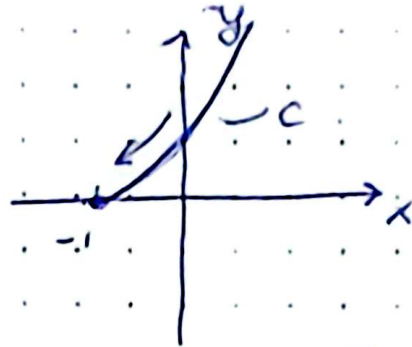
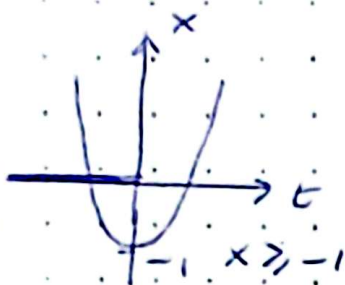
$$\sigma'(t) = (2\sin t, -2\cos t, 2\sin t)$$

$$\sigma'(0) = (0, -2, 0) \quad \|\sigma'(0)\| = 2$$

Cal 2 T2 MiMa.

1). $\bar{f}: (-\infty, 0] \rightarrow \mathbb{R}^2 \quad \bar{f}(t) = (t^2 - 1, t^4)$

$$\begin{cases} x = t^2 - 1 \rightarrow t^2 = x + 1 \\ y = t^4 \rightarrow y = (t^2)^2 = (x + 1)^2, x \geq -1 \end{cases}$$



$\bar{f}'(t) = (2t, 4t^3) \rightarrow$ huro. $t / \bar{f}(t) = (0, 1)$

$\begin{cases} t^2 - 1 = 0 \\ t^4 = 1 \end{cases} \rightarrow t = 1 \vee t = -1$ huro $1 \notin D\bar{f}$

$t = -1 \Rightarrow \bar{f}'(-1) = (-2, -4) = \vec{n}$

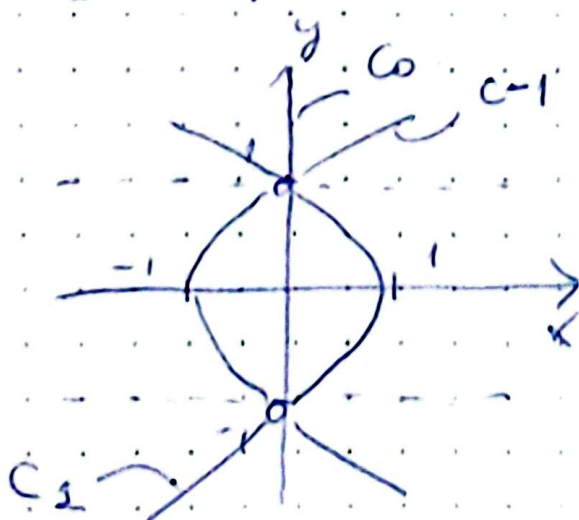
$\|\vec{n}\| = \sqrt{4 + 16} = \sqrt{20}$

4) $Df = \{(x, y) \in \mathbb{R}^2 / |y| \neq 1\} \quad y \neq 1 \wedge y \neq -1$

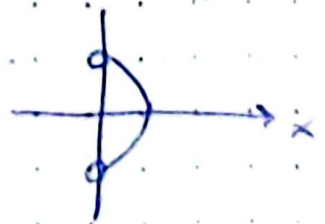
$C_0: x = 0$ en Df

$C_1: x = -y^2 + 1$ dentro de Df

$C_2: x = y^2 - 1$ " " " "



para medir el arco



$\sigma: (-1, 1) \rightarrow \mathbb{R}^2 /$

$\bar{f}(t) = (t^2 - 1, t^4)$



$$2) \quad T(x, y) = \cos(-x+2y) + e^{2-xy} + yx^2 + 1010$$

$$P_0 = (2, 1)$$

$$\nabla T(x, y) = (\sin(-x+2y) - y e^{2-xy} + 2xy, -2\sin(-x+2y) - x e^{2-xy} + x^2)$$

$$\nabla T(2, 1) = (0 - 1 + 4, 0 - 2 + 4) = (3, 2)$$

$$\|\nabla T(2, 1)\| = \sqrt{4+9} = \sqrt{13}$$

$$\vec{v}_{\min} = \left(-\frac{3}{\sqrt{13}}, -\frac{2}{\sqrt{13}} \right)$$

$$b) \quad \vec{v}_{\min_1} = \left(\frac{2}{\sqrt{13}}, \frac{-3}{\sqrt{13}} \right) \quad \text{y} \quad \vec{v}_{\min_2} = \left(\frac{-2}{\sqrt{13}}, \frac{3}{\sqrt{13}} \right)$$

$$3) \quad \begin{cases} j^2 + z^2 = 4 \\ j = x \end{cases}$$

$$a) \quad \varphi: [0, 2\pi] \rightarrow \mathbb{R}^3 / \varphi(t) = (2\cos t, 2\cos t, 2\sin t)$$

$$b) \quad \text{busca } t / \varphi(t) = (2, 2, 0) \rightarrow t = 0$$

$$\Rightarrow \varphi'(t) = (-2\sin t, -2\sin t, 2\cos t)$$

$$\varphi'(0) = (0, 0, 2)$$

$$\|\varphi'(0)\| = 2$$

